

Project Workflow: Week 7

Phase: Ping Pong Game Controls, Scoring & Game State

1. Objective

The primary goal of Week 7 was to improve the Ping Pong game by implementing player controls, score tracking, and basic game-state functionality to make the game more complete and interactive. This included:

- Implementing keyboard controls for both paddles.
- Adding score tracking for both players.
- Detecting when the ball goes out of bounds and assigning points.
- Creating pause and restart functionality.
- Testing the complete gameplay flow and responsiveness.

2. Detailed Task Breakdown

- Player Controls: Implemented keyboard-based controls using `KeyListener` so both players can move their paddles smoothly.
- Score Tracking: Added a scoring system to increase the correct player's score when the ball passes the opponent's side.
- Out-of-Bounds Detection: Implemented detection for when the ball leaves the left or right side of the game area.
- Pause & Restart: Added functionality to pause the game and restart the gameplay when required.
- Gameplay Testing: Tested paddle movement, scoring, ball movement, pause/restart behavior, and overall responsiveness.

3. Deliverables

- Functional keyboard controls for both players.
- Working score tracking system.
- Ball out-of-bounds and point assignment system.
- Pause and restart functionality.
- Tested and improved basic Ping Pong gameplay.

4. Tools & Resources

- Programming Language: Java
- GUI Framework: Java Swing
- Game Loop: `javax.swing.Timer`
- Keyboard Controls: `KeyListener`
- 2D Graphics: `java.awt.Graphics`
- Development Environment: IntelliJ IDEA / VS Code

- Reference Materials: Java Swing, KeyListener, Timer, score tracking, and game-state handling references